Problem 1

In order to test that the vectors produced by my code were actually eigenvectors of $\mathbf{A}$, I verified that for $\epsilon' \approx \epsilon$ given as a parameter

$$\sum_i \left((\mathbf{A}v)_i - \lambda v_i\right)^2 < \epsilon'$$

The following bash commands are an overview of how to reproduce the results that follow.

Figure 1 demonstrates the pattern of convergence given varialbes ϵ and N for matrix $\mathbf{A}$. As one might expect, the number of iterations required for the power method to converge is proportional to the selected precision. However it is also interesting to note that the number of iterations to convergence *decreases* as the matrix size increases.

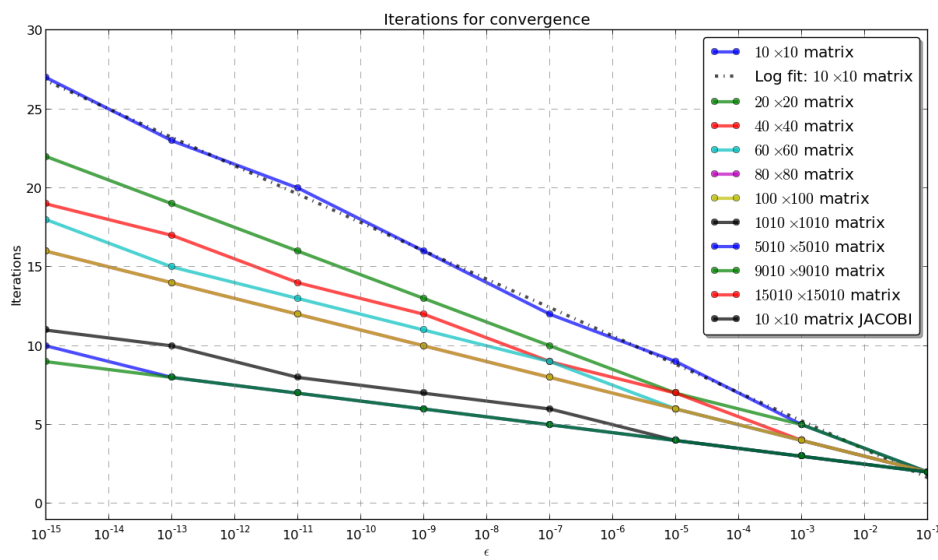

Figure 1: Plot of convergence iterations vs ϵ for selected matrix sizes

It can be noted that the number of iterations grew logarithmically with ϵ, i.e. for the 10×10 matrix,

$$\text{Iterations} \approx -0.78 * \log(\epsilon) - 0.16$$

with an $r^2 = .9989$. Furthermore, as a function of matrix size for a fized ϵ, the number of iteration necessary as a function of matrix size N was close to the following,

$$\text{Iterations} \approx 49.5 \times N^{-.43} + 9$$

Therefore, we can say that as a function of ϵ, convergence is on the order of $\mathcal{O}(\log(\epsilon))$.

Furthermore, I implemented the Jacobi algorithm. Although it was able to find the correct solution, it did using more iterative cycles. Given that each cycle required work to find the next pivot, and update the Jacobi matrix according, it also required more work per cycle. Therefore, the performance Jacobi implementation was below that of the power method.

Table 1: Jacobi Iterations

$\log_{10}(p)$	-1	-3	-5	-7	-9	-11
Iterations	109	126	141	154	162	168

Part 2

The margin of error for the reconstructed matrix falls quadratically as k increases. More specifically for a 10×20 matrix, the Frobenius norm followed the function

$$\text{Error} = y_1 \approx 2.48(k - 10.69)^2 - 9$$

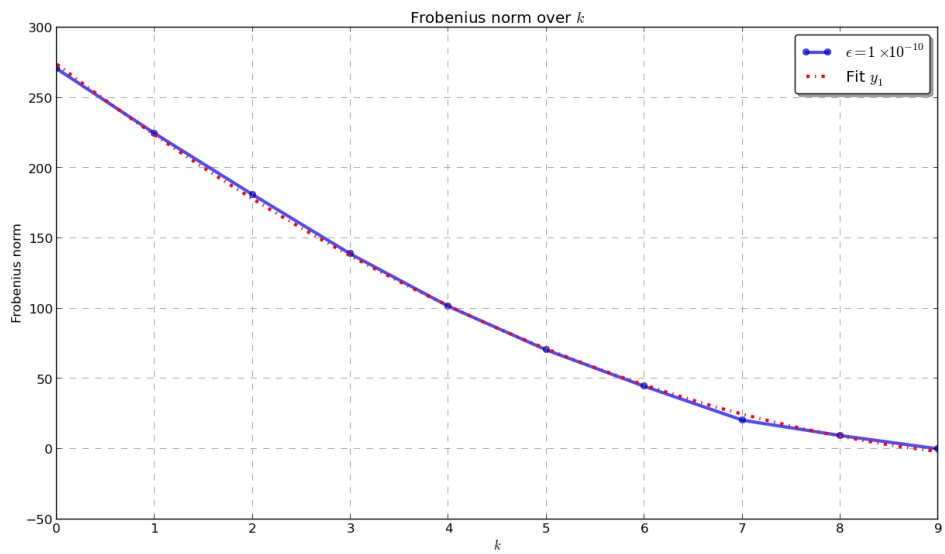

Figure 2: Plot of Frobenius norm vs k precision level $\epsilon = 1 \times 10^{-10}$

This is exactly as one might expect as use find more information we expect our model to be better. Since the eigenvectors are orthogonal, each new eigenvector used in reconstructing the matrix adds unique information.

Part 3

In Figure 3, we can see two trends, (1) a general trend that the error decreases as the probability that each element $\mathbf{A}_{ij}$ will be uncorrupted increases, the margin of error between the reconstructed matrix $\tilde{\mathbf{A}}$ decreases. Furthermore, (2) the error is smaller for smaller values of k.

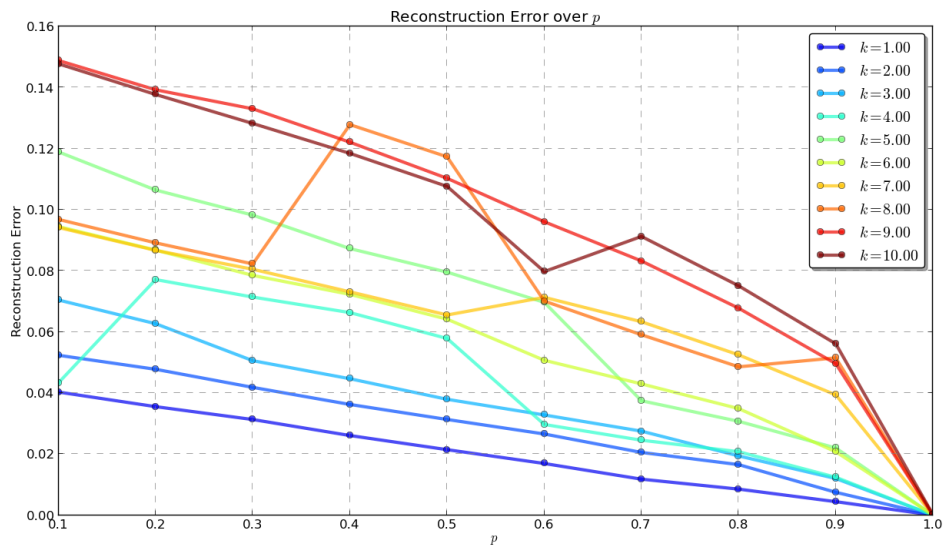

Figure 3: Plot of Frobenius norm vs k

At first it didn't make sense to me that the accuracy would decrease as k increased. This was because I was overlooking the fact that k was not only used to reconstruct the matrix, but also to determine its rank in the first place, and increasing, in a way, the entropy of the system.